Kepler-783

A candidate signal at 11.1458 days, in none of NASA's Kepler tables · KIC 10337517 · Kepler

Summary

The engine finds a dip of 87 parts per million in the light of Kepler-783 that repeats every 11.1458 days and lasts 4.6 hours. It was seen 102 times, at a signal-to-noise of 13.7.

The classifier puts the probability that this is a planet at 93.5%. If it is one, it is 0.88 times the size of Earth and orbits 0.093 AU from its star, where it would be at about 720 K.

This analysis was made after cutting out the transits of 2 other signals on the same star (at 7.0539 d, 4.2926 d), so that this one could be measured on its own.

The period matches none of the 2 objects NASA lists for this star.

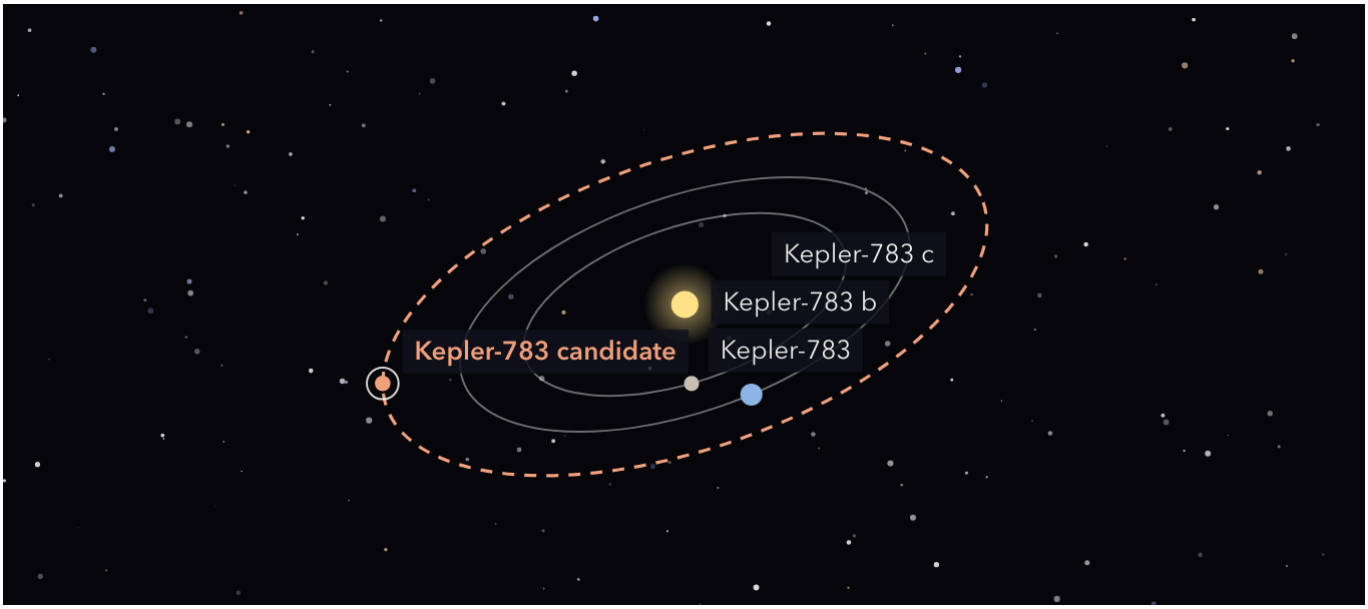
Of 8 automated tests, 0 count against a planet.

TRICERATOPS puts the false-positive probability at 7.39% (from nearby stars alone: 0.10%): not validated.

> Engine's verdict: **planet candidate**, above the 90% bar for human review. This is not a confirmation. A planet is confirmed by follow-up observation or by statistical validation, and this signal's validation result is "not validated".

Found in a blind survey as signal 3 of 3 on this star, at S/N 8.2 with a planet probability of 97.8%. With nothing there, the engine still reports a further signal on about 2 of every 105 stars.

The system · Orbits to scale ·



Body	Status	Size	Distance from the star	Its year	Temperature
Kepler-783 b	confirmed	0.78 R⊕	0.0496 AU	4.2926 d	991 K

Body	Status	Size	Distance from the star	Its year	Temperature
Kepler-783 c	confirmed	2.33 R⊕	0.0690 AU	7.0540 d	840 K
Kepler-783 candidate	candidate found by this engine	0.88 R⊕	0.0928 AU	11.1458 d	720 K

G-type star, 0.55 times the Sun's brightness, 1,683 light-years from Earth. Its habitable zone runs from 0.71 to 1.25 AU. Solid orbits are confirmed planets, dashed ones candidates; discs are enlarged.

Probability this is a planet

93.5%

S/N 13.7 · 102 transits

Planet candidate	93.49%
Eclipsing binary	6.30%
Stellar / instrument noise	0.21%

Five neural networks read the folded light curve; boosted trees weigh their answer with 57 measurements of the signal. The networks alone say 78.8% planet.

Period	Depth	Size if a planet
11.1458 d	87 ppm	0.88 R⊕

Duration of the dip · expected for this star	4.56 h · 3.72 h
Distance from the star	0.0928 AU · 23.1 star radii
Temperature there · sunlight against Earth's	720 K · 64×
Size, limb-darkened fit	0.79 ± 0.93 R⊕
Mid-transit time	142.1002 + n × 11.14579 BKJD
Search peak strength (SDE)	8.1

The star · Kepler-783 · KIC 10337517 · Kepler ·

Temperature	5,357 K
Radius · mass	0.863 R ⊙ · 0.859 M ⊙
Measurements used	46,138 over 1,459 days
Time between measurements	29.4 min
Scatter outside transit	176 ppm

NASA lists on this star	Period	Radius	Depth
Kepler-783 c confirmed	7.0539 d	2.36 R⊕	512 ppm
Kepler-783 b confirmed	4.2926 d	0.68 R⊕	61 ppm

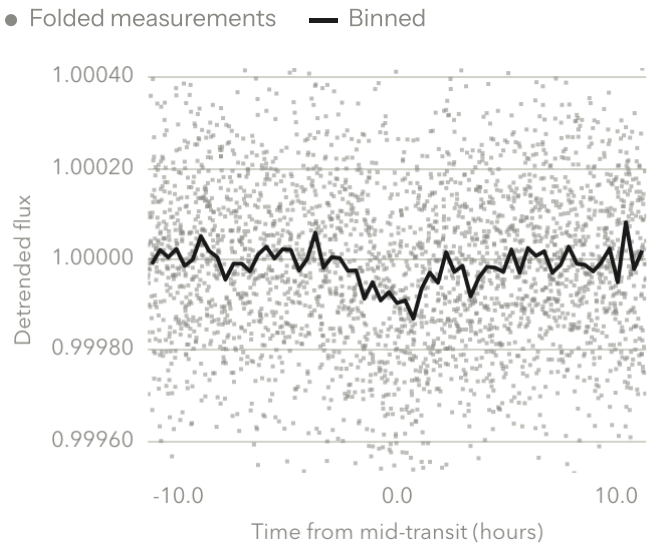
> This signal's period matches none of them.

Removed before this search: every transit of the signal at 7.0539 d (1.9 h each).

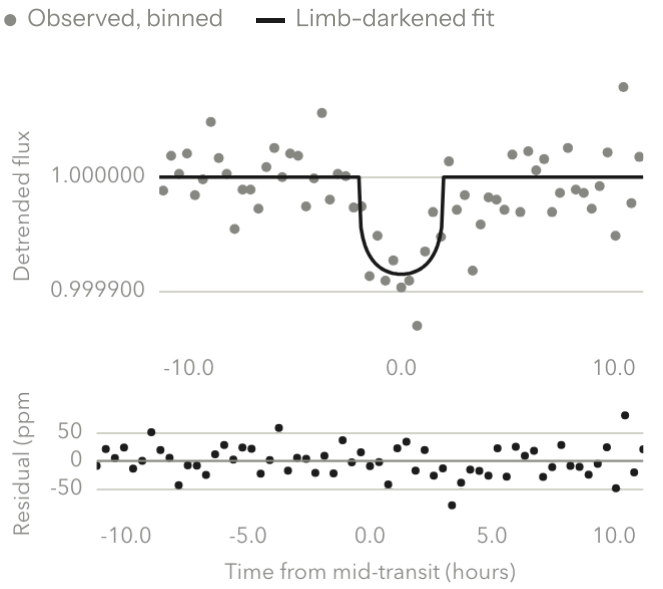
Removed before this search: every transit of the signal at 4.2926 d (2.1 h each).

The dip · Every transit folded on top of the others ·

Folded transit · measurements and the binned curve the classifier reads

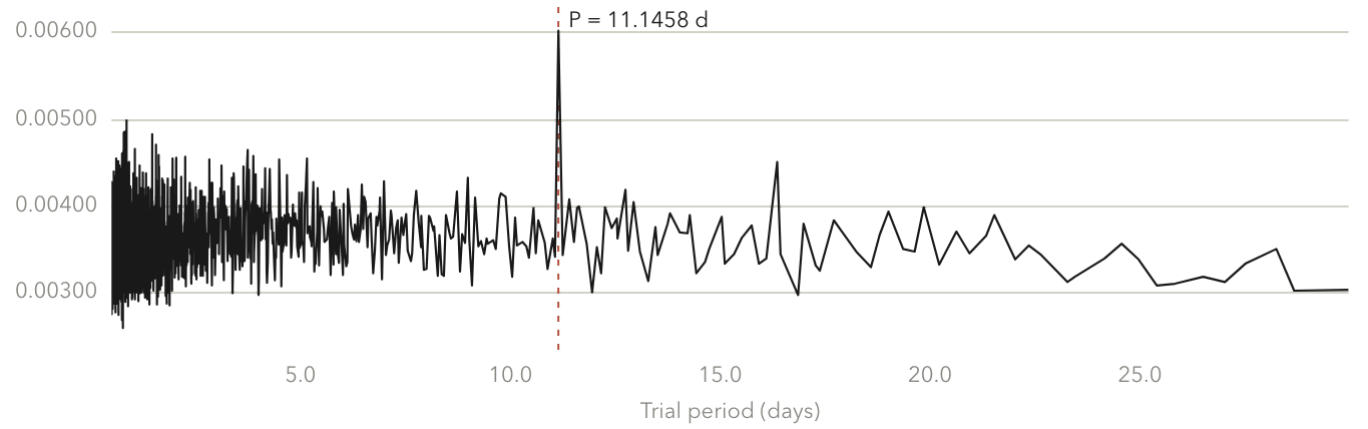


Fitted model against the binned data · residual scatter 28 ppm

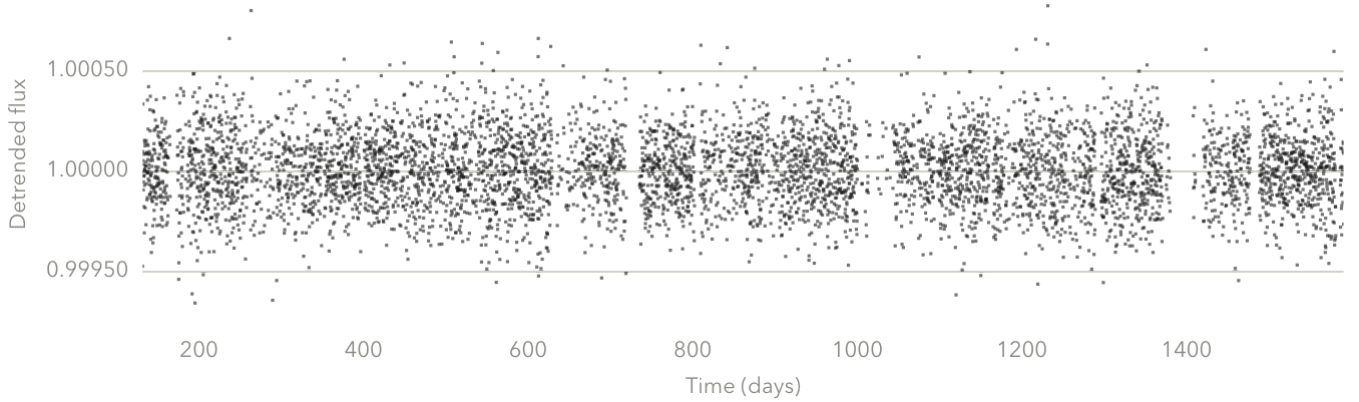


The search · 60,000 trial periods from 0.5 to 30 days ·

Strength of a repeating dip at each trial period

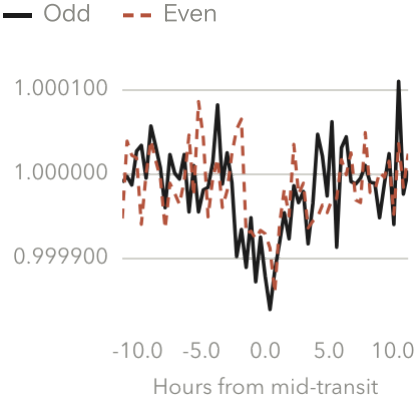


The light curve with the star's slow changes removed



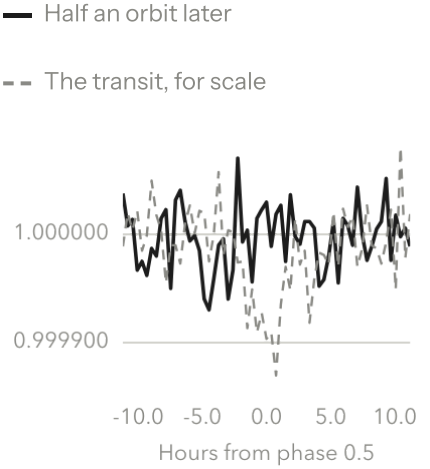
Is it something else? · Tests for eclipsing binaries, neighbouring stars and noise ·

Odd against even transits



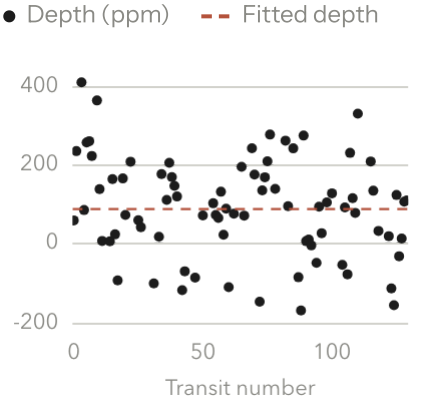
A planet gives identical curves.

Half an orbit later



A dip here is a second star's eclipse.

Depth of each transit



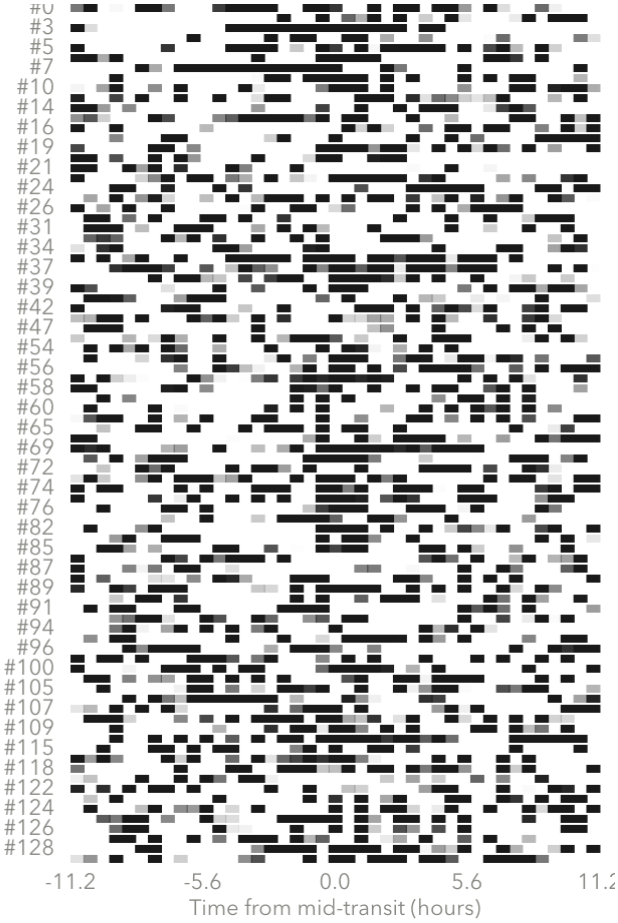
Dashed line is the fitted depth.

Test	Measured	Reading
Odd against even transits Two unequal stars eclipsing each other give alternating depths; a planet gives the same depth every time.	109 and 64 ppm, 2.5σ apart	consistent with a planet
Secondary eclipse, half an orbit later A dip here means the companion shines by its own light: a star, not a planet.	-13 ppm, -2.0σ	consistent with a planet
Movement of the star's image in transit If the image shifts during the dip, the dimming comes from a neighbouring star.	0.05 millipixels, 0.8σ	consistent with a planet

Test	Measured	Reading
Where on the detector the dip comes from The image in transit is subtracted from the image just outside it; the difference shows the source of the dip. Below about 150 ppm the difference image is noise-limited and the test has little power.	0.47 pixels from the target, 1.2σ, 7 quarters	no shift seen, but the test is weak at this depth
Duration against a transit across this star A dip much longer than the star allows points to a different, larger star; a much shorter one to a grazing eclipse.	1.23 of the expected 3.72 h	consistent with a planet
Shape of the dip Planets cross the whole disc and give flat-bottomed dips; grazing binaries give V shapes.	0.48 on a scale from 0 (flat-bottomed) to 1 (V-shaped)	consistent with a planet
Any other dip at this period A second significant dip at another phase is the other eclipse of a binary on an eccentric orbit.	strongest elsewhere 14 ppm at phase 0.78, 2.4σ	consistent with a planet
Size of the companion Anything much larger than Jupiter (11.2 Earth radii) is a star.	0.88 Earth radii	consistent with a planet

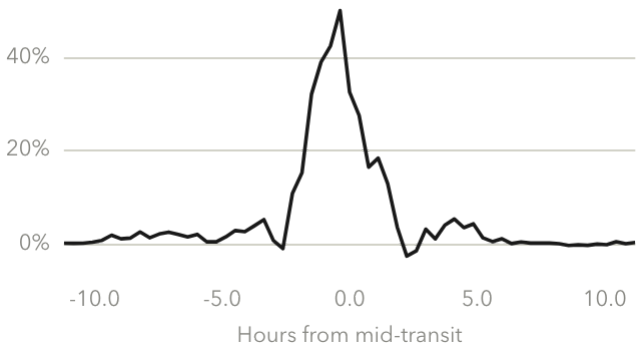
Transit by transit · 86 observed ·

Every transit, stacked



One row per transit; darker is deeper. A real signal lines up in a vertical band.

What the classifier relied on



How far the planet probability falls when that stretch of the folded curve is flattened.

Transit timing · Does each transit arrive on schedule? ·

> Not measurable for this signal: single transits are too shallow to time: each one stands only 1.5 times above the noise (at least 3 is needed). A signal this shallow only appears once all its transits are stacked, so there is no time to read off any single one.

Statistical validation · How likely is it to be something other than a planet? ·

False-positive probability

7.39 %

± 0.42 over 5 runs

From known nearby stars

0.10 %

± 0.01

> By the usual thresholds this is a **not validated** .
> Validated needs a false-positive probability under 1.5% with under 0.1% from nearby stars; likely planet needs under 50% with under 0.1%.

What could cause the dip	Probability
Planet on the target star TP · counts as a planet	72.09%
Planet on the target, which has an unseen bound companion PTP · counts as a planet	20.52%

The star already has 2 planets or candidates. False positives are rarer on such stars by a factor of about 50 (Lissauer et al. 2012), which would bring the false-positive probability to 0.15%. That adjustment does not change the part owed to nearby stars.

TRICERATOPS (Giacalone et al. 2021, AJ 161, 24), Kepler quarters 4, 5, 6, 7, 21 catalogued stars in the field, 1,000,000 simulated systems per scenario. No high-resolution imaging was used, so unseen close companions are limited only by the light curve's shape.

What could cause the dip	Probability
Planet on an unseen bound companion star STP	7.29%
Planet on a known nearby star NTP	0.06%
Eclipsing binary: a known nearby star NEB	0.04%
The same, at twice the period NEBx2P	0.01%

How this was made

Run	58746784e857
Analysed	11 Oct 2026, 13:59 UTC · 39.9 s
Data	Kepler quarters 1–17 · pdcsap_flux
Glitches removed	2 measurements beyond 5σ
Slow changes removed	Savitzky–Golay, 1.5 d window
Period search	Box least squares, 0.5–30 d

Classifier	5 neural networks + boosted trees
Its score on held-out Kepler stars (AUC)	0.997
Audit trail	14 records, hash-chained
Last record's hash	5b715131186de184…
Assumed for size and temperature	impact parameter 0.0 · albedo 0.3

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